ENGINEERING DRAFTING GUIDE

Electrical, Electronic, Logic, and Computer Engineering Schematics Using draw.io / diagrams.net

For Computer Engineering (CpE) and Electronics Engineering (ECE) Students

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Purpose of the Guide

This guide develops practical engineering drafting habits for schematic, logic, control, and embedded-system diagrams. It uses draw.io (diagrams.net) as the drafting platform and emphasizes legibility, electrical meaning, consistent symbols, professional labels, and clean documentation rather than decorative diagramming.

Source basis: the symbol families are adapted for instruction from the Electrical Technology reference page identified by the instructor, with draw.io procedures aligned to current official draw.io documentation. Symbol artwork in this guide is original instructional line art.

Contents

1. Learning Outcomes
2. Engineering Drafting for CpE and ECE
3. Getting Started with draw.io / diagrams.net
4. Core Schematic Drafting Rules
5. Electrical and Electronic Symbol Reference
6. Using Symbols and Connectors in draw.io
7. Reference Designators and Engineering Labels
8. Drafting Workflows for CpE and ECE
9. Guided Drafting Exercises
10. Student Submission Standard and Rubric
11. Quick Checklist
12. References

1. Learning Outcomes

At the end of the lesson or module, students should be able to:

- Identify and interpret common electrical, electronic, electromechanical, measurement, and digital-logic schematic symbols.
- Create a clean engineering schematic in draw.io using consistent symbols, orthogonal wiring, labels, and reference designators.
- Distinguish an electrical junction from a wire crossing with no electrical connection.
- Organize a schematic so signal flow, power distribution, control flow, and functional blocks are easy to follow.
- Apply suitable reference designators such as R1, C1, D1, Q1, U1, SW1, F1, and J1.
- Export a diagram at publication-ready quality for laboratory reports, design documents, research manuscripts, and capstone documentation.
- Evaluate a drawing for engineering readability, consistency, completeness, and traceability.

2. Engineering Drafting for CpE and ECE

Engineering drafting communicates how a system is organized and how its components are electrically or logically related. In CpE and ECE, a schematic is not merely a picture of physical placement. It is a technical abstraction that prioritizes function, connectivity, signal direction, power, control, interfaces, and component identity.

Key Principle

A schematic should answer three questions quickly: (1) What are the major components? (2) How are they connected? (3) What electrical or logical function does each connection perform?

2.1 Typical drawings students should be able to produce

- Basic electrical circuit schematics and DC/AC source-load diagrams.
- Analog electronic schematics: bias networks, op-amp circuits, filters, and sensor front ends.
- Digital logic diagrams: gates, latches, combinational logic, and simple control logic.
- Microcontroller and embedded-system interface schematics.
- Block diagrams showing processors, sensors, actuators, communication buses, and power domains.
- Power-supply and protection diagrams.
- Control and relay diagrams.
- Instrumentation diagrams showing meters, probes, signal sources, and measured nodes.

3. Getting Started with draw.io / diagrams.net

draw.io is the diagram editor associated with diagrams.net. The official documentation supports creating technical diagrams by adding shapes, connecting them with fixed or floating connectors, opening additional shape libraries, labeling shapes and connectors, styling elements, and exporting to formats such as PNG, SVG, and PDF.

3.1 Recommended classroom setup

1. Open the diagrams.net web editor or draw.io Desktop, then create a blank diagram.

2. Choose a storage location appropriate for the class: Device, Drive, OneDrive, or another approved course location.

3. Set the page orientation to Landscape for wide schematics when necessary.
4. Keep the grid and snap-to-grid features enabled for alignment.
5. Open More Shapes at the bottom of the left panel and enable the electrical, logic, or other engineering-related shape libraries available in the installed version.
6. Keep the General library available because the Misc group contains the waypoint shape used for visible connection points.
7. Use the Search Shapes field when a component is difficult to locate.
8. Save the editable source file in .drawio format before exporting an image or PDF.

Instructor Demonstration

Show students how to drag a shape, resize it, rotate only when electrically meaningful, connect it with a connector, double-click to label it, align multiple components, copy/paste repeated symbols, and export the final page.

4. Core Schematic Drafting Rules

Rule 1 - Draw for function, not physical appearance: Use schematic symbols and functional blocks. Do not draw a photorealistic resistor, Arduino board, relay, or sensor unless the purpose is a wiring/installation diagram.

Rule 2 - Maintain a dominant flow direction: For most schematics, place inputs or sources toward the left/top and outputs or loads toward the right/bottom. Keep related signal paths visually consistent.

Rule 3 - Prefer horizontal and vertical wiring: Use orthogonal routes. Avoid unnecessary diagonals, curved wires, and decorative routing.

Rule 4 - Minimize wire crossings: Reposition components or use net labels before allowing many crossing conductors. Every crossing makes interpretation harder.

Rule 5 - Show junctions explicitly: Use a filled junction dot/waypoint when conductors are electrically connected. For wires that only cross, do not add a junction dot; use line jumps where needed.

Rule 6 - Use consistent symbol conventions: Do not mix multiple symbol standards randomly in the same drawing. Choose the symbol style used in the course or required by the project.

Rule 7 - Label components and values: A resistor should not remain simply “resistor.” Give it an identifier such as R1 and, when known, a value such as 1 kΩ.

Rule 8 - Name important nets: Label power rails and important signals: +5V, +3V3, +12V, GND, SDA, SCL, TX, RX, PWM, RESET, ADC_IN, etc.

Rule 9 - Keep power visually organized: Place positive supply rails toward the top and common/ground references toward the bottom when this improves readability.

Rule 10 - Group functional stages: Separate power, sensing, processing, communication, driver, and load sections. Use headings or light boundaries only when they improve interpretation.

Rule 11 - Avoid dangling or ambiguous connections: Every wire end should clearly terminate at a component pin, connector, junction, or labeled off-page/net reference.

Rule 12 - Preserve engineering traceability: Use reference designators, pin numbers or names, connector identifiers, and signal labels that correspond to the bill of materials, code, PCB, or test procedure where applicable.

5. Electrical and Electronic Symbol Reference

The following symbol set is intended for introductory classroom drafting. Symbol appearance can vary by IEC, ANSI/IEEE, manufacturer library, and application. Students should learn the meaning of the symbol and then use one consistent convention within a drawing.



Resistor

Typical reference designator: R

Purpose: Limits current, divides voltage, and biases active devices.

draw.io search term: resistor

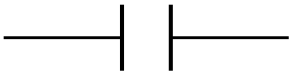


Variable Resistor / Potentiometer

Typical reference designator: VR / RV

Purpose: Provides adjustable resistance or voltage division.

draw.io search term: variable resistor

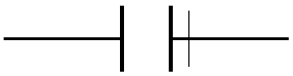


Capacitor

Typical reference designator: C

Purpose: Stores electric-field energy; used for filtering, coupling, timing, and decoupling.

draw.io search term: capacitor



Polarized Capacitor

Typical reference designator: C

Purpose: Capacitor with polarity; observe the marked positive/negative terminal.

draw.io search term: polarized capacitor

Inductor

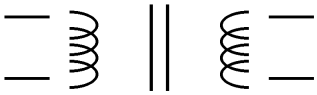


Typical reference designator: L

Purpose: Stores magnetic-field energy; used in filters, chokes, converters, and RF circuits.

draw.io search term: inductor

Transformer

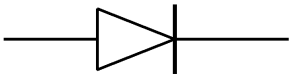


Typical reference designator: T

Purpose: Transfers AC energy magnetically and may step voltage up/down or provide isolation.

draw.io search term: transformer

Diode

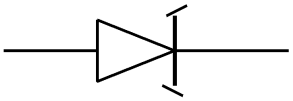


Typical reference designator: D

Purpose: Conducts primarily in one direction; commonly used for rectification and protection.

draw.io search term: diode

Zener Diode

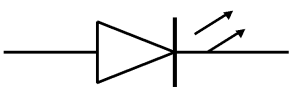


Typical reference designator: ZD / D

Purpose: Used for voltage reference, clamping, and regulation in reverse breakdown.

draw.io search term: zener

Light-Emitting Diode (LED)

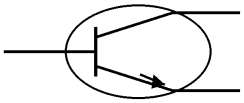


Typical reference designator: LED / D

Purpose: Emits light when forward biased; arrows point outward from the diode.

draw.io search term: led

NPN BJT

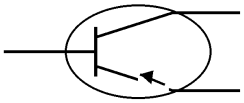


Typical reference designator: Q

Purpose: Three-terminal current-controlled transistor used for switching or amplification.

draw.io search term: npn transistor

PNP BJT

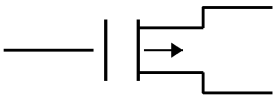


Typical reference designator: Q

Purpose: Complementary BJT; emitter arrow points inward toward the base.

draw.io search term: pnp transistor

N-Channel MOSFET

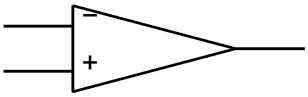


Typical reference designator: Q

Purpose: Voltage-controlled transistor widely used for switching and amplification.

draw.io search term: nmos

Operational Amplifier



Typical reference designator: U

Purpose: High-gain differential amplifier block with inverting and non-inverting inputs.

draw.io search term: op amp

Battery

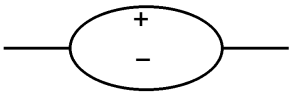


Typical reference designator: BT / B

Purpose: DC electrochemical source; long/short plates indicate polarity in common notation.

draw.io search term: battery

DC Source



Typical reference designator: V

Purpose: Independent direct-voltage source.

draw.io search term: dc source

AC Source



Typical reference designator: V

Purpose: Independent alternating-voltage source.

draw.io search term: ac source

Earth / Reference Ground

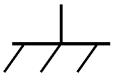


Typical reference designator: GND

Purpose: Common electrical reference or return; use the correct ground type for the design.

draw.io search term: ground

Chassis Ground



Typical reference designator: GND

Purpose: Represents a conductive chassis or enclosure reference.

draw.io search term: chassis ground

SPST Switch



Typical reference designator: SW

Purpose: Single-pole single-throw switch used to make or break one circuit path.

draw.io search term: switch

Push Button

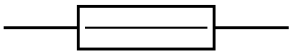


Typical reference designator: PB / SW

Purpose: Momentary switch operated by pressing a button.

draw.io search term: push button

Fuse

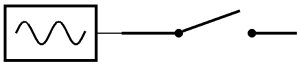


Typical reference designator: F

Purpose: Overcurrent protection element designed to open when current exceeds its rating.

draw.io search term: fuse

Relay

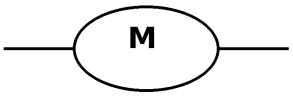


Typical reference designator: K / RY

Purpose: Electrically operated switch that separates a control circuit from switched contacts.

draw.io search term: relay

Electric Motor

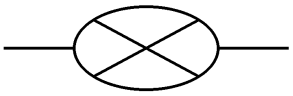


Typical reference designator: M

Purpose: Converts electrical energy to mechanical motion.

draw.io search term: motor

Lamp / Indicator

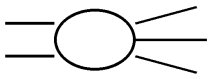


Typical reference designator: LAMP

Purpose: Converts electrical energy to light; useful for load and indicator schematics.

draw.io search term: lamp

Microphone

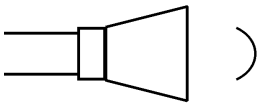


Typical reference designator: MIC

Purpose: Converts acoustic energy to an electrical signal.

draw.io search term: microphone

Loudspeaker

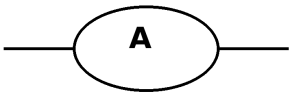


Typical reference designator: SPK

Purpose: Converts an electrical signal to acoustic output.

draw.io search term: speaker

Ammeter

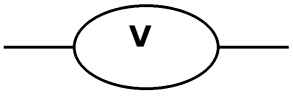


Typical reference designator: A

Purpose: Measures current and is inserted in series with the measured branch.

draw.io search term: ammeter

Voltmeter

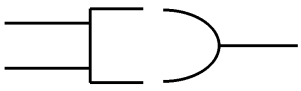


Typical reference designator: V

Purpose: Measures potential difference and is connected in parallel across two points.

draw.io search term: voltmeter

AND Gate



Typical reference designator: U / G

Purpose: Output is HIGH only when all required inputs are HIGH.

draw.io search term: and gate

OR Gate

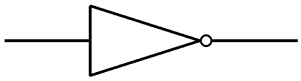


Typical reference designator: U / G

Purpose: Output is HIGH when one or more inputs are HIGH.

draw.io search term: or gate

NOT Gate / Inverter



Typical reference designator: U / G

Purpose: Produces the logical complement of its input.

draw.io search term: not gate

NAND Gate



Typical reference designator: U / G

Purpose: Inverted AND function.

draw.io search term: nand gate

NOR Gate



Typical reference designator: U / G

Purpose: Inverted OR function.

draw.io search term: nor gate

XOR Gate



Typical reference designator: U / G

Purpose: Output is HIGH when inputs differ for the common two-input form.

draw.io search term: xor gate

7-Segment Display

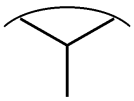


Typical reference designator: DS

Purpose: Seven individually controlled segments used to display numerals and limited characters.

draw.io search term: 7 segment

Antenna

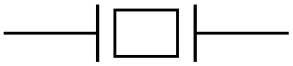


Typical reference designator: ANT

Purpose: Interface between guided electrical signals and radiated electromagnetic waves.

draw.io search term: antenna

Crystal / Resonator

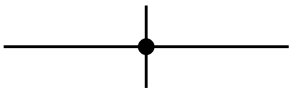


Typical reference designator: Y / X

Purpose: Frequency-selective resonator commonly used for clocks and oscillators.

draw.io search term: crystal

Connected Junction

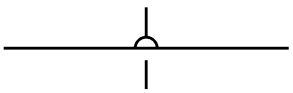


Typical reference designator: —

Purpose: Filled node indicates that conductors are electrically connected.

draw.io search term: waypoint

Wire Crossing - No Connection



Typical reference designator: —

Purpose: Crossing conductors without a junction dot are not connected; a line jump can clarify this.

draw.io search term: line jump

6. Using Symbols and Connectors in draw.io

6.1 Place and search for symbols

1. Open the appropriate shape libraries using More Shapes, then click Apply.

2. Drag the selected symbol onto the canvas.

3. Use the shape search field if the symbol is not immediately visible.

4. Keep repeated components the same size. Copy an existing component when possible rather than resizing every instance independently.

5. Use Arrange > Align and Distribute to establish consistent rows and spacing.

6.2 Connect components correctly

1. Use connectors rather than ordinary decorative lines so the wire remains attached when components move.

2. Prefer horizontal/vertical connector paths for most schematics.

3. Use a waypoint shape at a true electrical junction. The draw.io documentation specifically recommends waypoints for visible contact points in circuit and logic diagrams.

4. For crossing wires that are not connected, avoid a waypoint and enable line jumps on one connector when the crossing could be misunderstood.

5. Re-route crowded wires instead of stacking several lines directly over one another.

6.3 Labels and typography

- Use a single readable font family throughout the schematic.
- Use approximately 9-12 pt equivalent text for normal labels and larger text for functional headings.
- Place reference designators close to their symbols without touching wires.
- Use engineering units and prefixes correctly: Ω, kΩ, MΩ, pF, nF, µF, mH, V, mA, MHz.
- Do not use color as the only way to communicate a circuit meaning; the diagram should remain interpretable in black and white.

7. Reference Designators and Engineering Labels

Designator	Common Component	Example	Recommended Label
R	Resistor	R1	R1 10 kΩ
C	Capacitor	C3	C3 100 nF
L	Inductor	L1	L1 10 µH
D	Diode	D2	D2 1N4148
LED	LED	LED1	LED1 RED
Q	Transistor / MOSFET	Q1	Q1 2N3904
U	Integrated circuit	U1	U1 LM358
SW	Switch	SW1	SW1 RESET
F	Fuse	F1	F1 1 A
K / RY	Relay	K1	K1 5 V Relay
J / P	Connector	J1	J1 SENSOR IN
BT	Battery	BT1	BT1 9 V
M	Motor	M1	M1 12 V DC
Y / X	Crystal	Y1	Y1 16 MHz
TP	Test point	TP1	TP1 ADC_IN

Reference Designator Rule

Each component instance must have a unique identifier. Do not label three resistors as R1. Use R1, R2, R3, etc. If an integrated circuit contains multiple units, a common convention is U1A, U1B, and so on where supported by the schematic style.

8. Drafting Workflows for CpE and ECE

8.1 CpE workflow - embedded controller interface

1. Draw the regulated power rails and ground reference.

2. Place the microcontroller or processor block at the center.

3. Place sensors and input devices on the left; place actuators, drivers, or outputs on the right.

4. Label digital and analog signals with meaningful net names.

5. Identify communication buses such as UART, I2C, SPI, CAN, or USB and show direction where helpful.

6. Add interface components such as pull-up resistors, current-limiting resistors, transistor drivers, level shifters, connectors, and protection devices.

7. Add test points for signals students are expected to measure.

8. Check that all supply pins, grounds, and required enable/reset signals are accounted for.

8.2 ECE workflow - analog / mixed-signal circuit

1. Start with the signal source or sensor at the left.

2. Arrange conditioning stages in sequence: protection, biasing, filtering, amplification, conversion, and output.

3. Keep feedback paths short and visually distinct.

4. Label component values, op-amp sections, transistor types, and supply rails.

5. Place decoupling capacitors near the associated IC in the logical schematic grouping.

6. Indicate important measurement nodes and expected signal names.

7. Separate sensitive analog ground references from noisy power paths when the design concept requires it, and clearly document how they join.

8.3 Logic diagram workflow

1. Write the Boolean requirement or truth-table behavior before drawing.

2. Place input signals on the left and output signals on the right.

3. Use gates with consistent orientation and equal spacing.

4. Label intermediate nets when the logic has more than one stage.

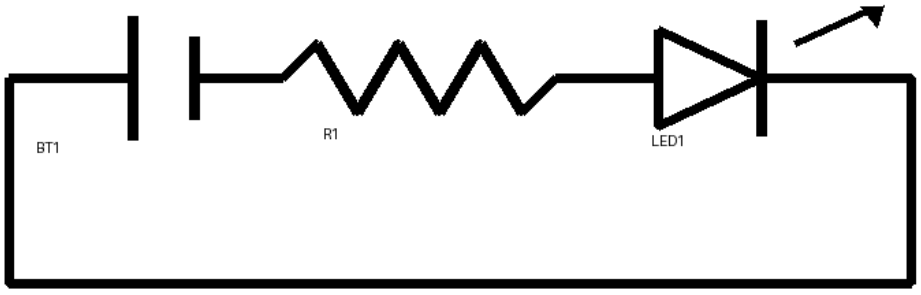
5. Use junction dots only where logic lines truly branch.

6. Validate the final drawing against the Boolean expression or truth table.

9. Guided Drafting Exercises

Exercise 1 - Battery, resistor, and LED series circuit

Objective: practice source, resistor, LED, connector routing, polarity, reference designators, and component values.



- Draft the circuit in draw.io.
- Use BT1, R1, and LED1.
- Assign BT1 = 9 V and R1 = 1 kΩ.
- Show the current path clearly.
- Add a title and student identification in a small title block.

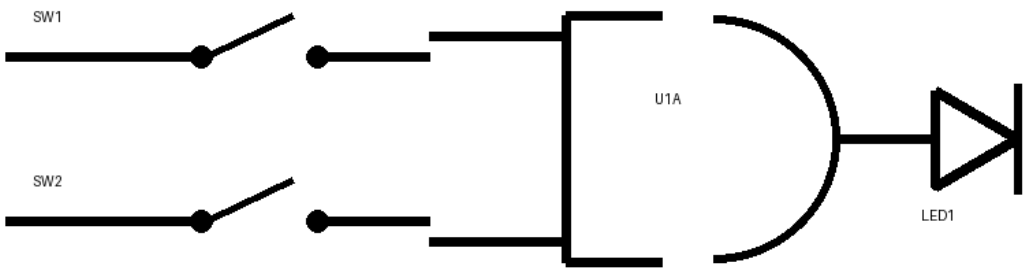
Exercise 2 - Voltage-divider measurement schematic

- Create a 5 V source feeding R1 = 10 kΩ and R2 = 10 kΩ in series.
- Label the midpoint VOUT.
- Place a voltmeter across VOUT and GND.
- Add a junction dot at the divider midpoint.
- Explain why the voltmeter is drawn in parallel, not series.

Exercise 3 - Transistor relay driver

- Use one NPN transistor, one base resistor, one relay coil, one flyback diode, one push button or logic input, and a DC supply.
- Label the transistor Q1, diode D1, resistor R1, and relay K1.
- Show coil suppression diode polarity correctly.
- Keep the control side visually separate from the switched load contact.

Exercise 4 - Two-input logic function



- Draft a two-input AND function with switches SW1 and SW2 and output LED1.
- Add a current-limiting resistor for the LED in your completed circuit-level version.
- Create a second page implementing the same function using NAND gates only.

Exercise 5 - CpE embedded-system block schematic

- Create a diagram containing: 5 V input, 3.3 V regulator, microcontroller, temperature sensor, push button, status LED, I2C OLED display, UART connector, and ground.
- Label SDA, SCL, TX, RX, RESET, +5V, +3V3, and GND nets.
- Include pull-up resistors on SDA and SCL.
- Organize the drawing into Power, Inputs, Processing, Communications, and Outputs.

Exercise 6 - ECE signal-conditioning schematic

- Create a sensor input followed by an RC low-pass filter, op-amp non-inverting amplifier, output test point, and ADC interface connector.
- Use appropriate R and C reference designators and values selected by the instructor.
- Label VIN, VFILT, VAMP, ADC_IN, +5V, and GND.
- Keep the signal path left-to-right and feedback path compact.

10. Student Submission Standard and Rubric

Recommended submission package: one editable .drawio file plus one exported PDF or high-resolution PNG. The drawing should contain a title, student name/group, course/section, date, and drawing revision.

Criterion	Excellent	Acceptable	Needs Improvement
Symbol correctness - 25%	Correct, consistent symbols and polarity.	Minor symbol inconsistency without changing meaning.	Incorrect or ambiguous symbols; polarity errors.
Connectivity - 25%	All connections clear; junctions and crossings unambiguous.	A few crowded routes but electrical meaning remains clear.	Dangling wires, unclear junctions, or incorrect connections.
Organization and readability - 20%	Strong flow, alignment, spacing, and functional grouping.	Generally readable with limited clutter.	Crowded, inconsistent, or difficult to follow.
Labels and traceability - 20%	Complete designators, values, signal names, and meaningful title information.	Most labels complete; minor omissions.	Missing designators/values/net names.
File quality and submission - 10%	Editable file plus clean export; no clipping or artifacts.	Files open correctly with minor layout issues.	Wrong file type, clipping, low resolution, or missing source file.

11. Quick Checklist Before Submission

☐ All components use correct and consistent schematic symbols.

☐ Each component has a unique reference designator.

☐ Important component values or part numbers are shown.

☐ Power rails and ground references are clearly labeled.

☐ Important signal nets are named.

☐ Wires are primarily horizontal or vertical and are not unnecessarily tangled.

☐ Electrical junctions are shown with clear connection dots/waypoints.

☐ Wire crossings without connection are visually unambiguous.

☐ There are no unintended dangling connectors.

☐ Symbols are aligned and consistently sized.

☐ Text does not overlap symbols or wires.

☐ The schematic has a clear title and revision/student information.

☐ The editable .drawio source is saved.

☐ The exported image/PDF is readable at normal zoom and has no clipping.

12. References

Electrical Technology. “Basic Electrical and Electronic Symbols of EE Components & Elements.” Reference page supplied by the instructor. https://www.electricaltechnology.org/2019/09/basic-electrical-electronic-symbols.html

draw.io. “Introduction - draw.io User Manual.” https://www.drawio.com/docs/manual/

draw.io. “Join Connectors with the Waypoint Shape.” https://www.drawio.com/docs/manual/shapes/waypoint-shape/

draw.io. “Example Technical Diagrams.” https://www.drawio.com/docs/diagram-types/

Teaching Note

The online reference page is useful for introducing symbol meaning, but professional projects may require a specific organizational, IEC, IEEE/ANSI, manufacturer, or regulatory symbol convention. Require students to state the adopted convention when a project is assessed against a formal standard.